Aggregate Function & Grouping Exercise 2

ID	First_name	Last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sara	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Hall	Laura	IT	50000	2020-08-10	San Francisco

EMPLOYEES EXCERCISE

① QUERY: COUNT () THE NUMBER OF EMPLOYEES

```
# SELECT COUNT (ID) AS NUMBER_EMPLOYEES
  FROM `WORKPLACE`.`DEFAULT`.`EMPLOYEES_1`;
```

NUMBER_EMPLOYEES
10

② TOTAL SALAD PAID TO EMPLOYEES IN THE IT DEPARTMENT

```
# QUERY: SELECT SUM (SALARY) AS TOTAL_SALARY
  FROM `WORKSPACE`.`DEFAULT`.`EMPLOYEES_1`
  WHERE DEPARTMENT = 'IT';
```

TOTAL_SALARY
22 00 00

③ QUERY: CALCULATE THE AVERAGE SALARY IN THE HR DEPARTMENT

```
# SELECT AVG (SALARY) AS AVERAGE_SALARY
  FROM `WORKSPACE`.`DEFAULT`.`EMPLOYEES_1`
  WHERE DEPARTMENT = 'HR';
```

AVERAGE_SALARY
49500

④ QUERY: FIND THE HIGHEST AND LOWEST SALARY IN THE COMPANY

```
# SELECT MAX (SALARY) AS HIGHEST_SALARY,
         MIN (SALARY) AS LOWEST_SALARY
  FROM `WORKSPACE`.`DEFAULT`.`EMPLOYEES_1`;
```

HIGHEST_SALARY	LOWEST_SALARY
62 000	48 000

⑤ QUERY: GROUP EMPLOYEES BY DEPARTMENT AND TOTAL SALARY IN EACH DEPARTMENT

```
# SELECT DEPARTMENT,
        SUM (SALARY) AS TOTAL_SALARY
  FROM 'WORKSPACE'.'DEFAULT'.'EMPLOYEES_1'
  GROUP BY DEPARTMENT;
```

DEPARTMENT	TOTAL_SALARY
IT	220000
HR	99000
FINANCE	119000
MARKETING	105000

⑥ QUERY: COUNT HOW MANY EMPLOYEES IN EACH CITY

```
# SELECT CITY,
  COUNT (ID) AS NUMBER_EMPLOYEES
  FROM 'WORKSPACE', 'DEFAULT'.'EMPLOYEES_1'
  GROUP CITY;
```

CITY	NUMBER_EMPLOYEES
NEW YORK	2
CHICAGO	3
LOS ANGELES	2
SAN FRANCISCO	2
HOUSTON	1

(8) QUERY : GROUP EMPLOYEES BY DEPARTMENT AND CALCULATE AVG SALARY IN THE (EACH) DEPARTMENT, ORDER THE RESULTS IN DECENDING ORDER

```
# SELECT DEPARTMENT,
        AVG (SALARY) AS AVERAGE_SALARY
FROM 'WORKSPACE'.'DEFAULT'.'EMPLOYEES_1'
GROUP BY DEPARTMENT
ORDER BY SALARY DESC;
```

DEPARTMENT	AVERAGE_SALARY
FINANCE	59500
IT	55000
MARKETING	52500
HR	49500

(7) QUERY : FIND THE DEPARTMENT WHERE THE TOTAL SALARY PAID EXCEEDS 100 000 (HAVING CLAUSE)

```
# SELECT DEPARTMENT
        SUM(SALARY) AS TOTAL_SALARY
FROM 'WORKSPACE'.'DEFAULT'.'EMPLOYEES_1'
GROUP BY DEPARTMENT
HAVING TOTAL_SALARY > 100 000;
```

DEPARTMENT	TOTAL_SALARY
IT	228 000
FINANCE	119 000
MARKETING	105 000

(9) LIST CITIES WHERE MORE THAN ONE EMPLOYEE WORKS, ORDERED BY THE NUMBER OF EMPLOYEES DESCENDING ORDER

```
# SELECT CITY,
        COUNT(ID) AS NUMBER_OF_EMPLOYEES
FROM 'WORKSPACE'.'DEFAULT'.'EMPLOYEES_1'
HAVING NUMBER_OF_EMPLOYEES > 1
ORDER BY NUMBER_OF_EMPLOYEES DESC;
```

CITY	NUMBER_OF_EMPLOYEES
CHICAGO	3
NEW YORK	2
LOS ANGELES	2
SAN FRANCISCO	2

(10) QUERY: FIND THE DEPARTMENT WITH THE HIGHEST AVERAGE SALARY

```
# SELECT DEPARTMENT
        AVG(SALARY) AS AVERAGE_SALARY
FROM 'WORKSPACE'.'DEFAULT'.'EMPLOYEES_1'
GROUP BY DEPARTMENT
ORDER BY AVERAGE_SALARY DESC
LIMIT 1;
```

DEPARTMENT	AVERAGE_SALARY
FINANCE	59500

(11) RETRIEVE EMPLOYEES FROM NEW YORK OR TORONTO, ORDERED BY SALARY DESCENDING

QUERY RETURN: EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY, CITY

```
# SELECT EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY, CITY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
WHERE CITY IN ('NEW YORK', 'TORONTO')
ORDER BY SALARY DESC;
```

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPARTMENT	SALARY	CITY
6	MICHAEL	CLARK	FINANCE	80000	NEW YORK
3	BOB	JOHNSON	FINANCE	75000	TORONTO
1	JOHN	DOE	IT	70000	NEW YORK

(12) RETRIEVE THE TOP 3 HIGHEST-PAID EMPLOYEES

QUERY RETURN: EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY

```
# SELECT EMPLOYEEID, FIRSTNAME, LASTNAME, DEPARTMENT, SALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
ORDER BY SALARY DESC
LIMIT 3;
```

EMPLOYEEID	FIRSTNAME	LASTNAME	DEPARTMENT	SALARY
6	MICHAEL	CLARK	FINANCE	80000
3	BOB	JOHNSON	FINANCE	75000
4	DAVID	BROWN	IT	72000

(13) FIND TOTAL SALARY PER DEPARTMENT, SORT IN DESCENDING ORDER

QUERY RETURN: DEPARTMENT, TOTALSALARY

```
# SELECT DEPARTMENT, SUM(SALARY) AS TOTALSALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
GROUP BY DEPARTMENT
ORDER BY TOTALSALARY DESC;
```

DEPARTMENT	TOTALSALARY
FINANCE	155000
IT	142000
SALES	65000
HR	60000

(14) FIND THE AVERAGE SALARY PER CITY, BUT ONLY INCLUDE CITIES WHERE THE AVERAGE SALARY IS GREATER THAN 65000

QUERY RETURN: AVERAGESALARY (RENAME AVG(SALARY) AS AVERAGESALARY)

```
# SELECT CITY,
        AVG(SALARY) AS AVERAGESALARY
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
GROUP BY CITY
HAVING AVERAGESALARY > 65000;
```

CITY	AVERAGESALARY
NEW YORK	75000
TORONTO	75000
LONDON	72000

(15) COUNT THE NUMBER OF EMPLOYEES PER DEPARTMENT, INCLUDING ONLY DEPARTMENTS WITH MORE THAN 1 EMPLOYEE

QUERY RETURN : EMPLOYEECOUNT (RENAME COUNT(EMPLOYEE ID) AS EMPLOYEECOUNT)

```
# SELECT DEPARTMENT,
        COUNT(EMPLOYEE ID) AS EMPLOYEE COUNT
FROM `WORKSPACE`.`DEFAULT`.`TABLE_1_EMPLOYEES_SQL_JOIN`
GROUP BY DEPARTMENT
HAVING EMPLOYEEMENT > 1;
```

DEPARTMENT	EMPLOYEE COUNT
IT	2
FINANCE	2

(16) RETRIEVE THE NUMBER OF PROJECTS PER STATUS, BUT ONLY INCLUDE STATUSES WITH AT LEAST 2 PROJECTS.

QUERY RETURN : STATUS, PROJECT COUNT (RENAME COUNT (PROJECT ID) AS PROJECT COUNT)

```
# SELECT STATUS, COUNT(PROJECT ID) AS PROJECT COUNT
FROM `WORKSPACE`.`DEFAULT`.`TABLE_2_PROJECTS_SQL_JOIN`
GROUP BY STATUS
HAVING PROJECT COUNT >= 2;
```

STATUS	PROJECT COUNT
COMPLETED	2
ONGOING	2
PENDING	2

(17) RETRIEVING THE TOTAL PROJECT BUDGET PER EMPLOYEE, BUT ONLY FOR EMPLOYEES WHO ARE MANAGING PROJECTS WORTH MORE THAN 150 000

QUERY RETURN: EMPLOYEE ID, FIRST NAME, LAST NAME, TOTAL PROJECT BUDGET (RENAME SUM(BUDGET) AS TOTAL PROJECT BUDGET)

```
# SELECT A.EMPLOYEEID, A.FIRSTNAME, A.LASTNAME,
        SUM(BUDGET) AS TOTALPROJECTBUDGET
FROM 'WORKSPACE'.'DEFAULT'.'TABLE_1_EMPLOYEES_SQL_JOIN' AS A
INNER JOIN 'WORKSPACE'.'DEFAULT'.'TABLE_2_PROJECTS_SQL_JOIN' AS B
ON A.EMPLOYEE ID = B.EMPLOYEE ID
GROUP BY A.EMPLOYEE ID, A.FIRSTNAME, A.LASTNAME
HAVING TOTAL PROJECT BUDGET > 150000,
```

EMPLOYEE ID	FIRST NAME	LAST NAME	TOTAL PROJECT BUDGET
1	JOHN	DOE	175000